

“Empowering Science and Data Analytics through Python”

Lecture_01

by

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Topic:

- 1. Operators and Expressions,**
- 2. Arithmetic/logic operators,**
- 3. Type conversion,**
- 4. F-strings,**
- 5. Intro to debugging.**

1. Operators and Expressions

What is an Operator?

Definition:

An **operator** is symbol used to perform an operation on one or more values (called operands).

Example:

Operands operator Operands $\rightarrow 3 + 5$

What is an Expression in programming language?

Definition: An expression in a programming language is any combination of values, variables, operators, and function calls that produces a result.

Example:

1. Arithmetic Expression: $5 + 3$,
2. Using Variables: $x = 10$, $x * 2$
3. Logical Expression: $5 > 3$
4. Function expression: `len("Hello")`

Expression Vs statement:

1. Expression: gives a value: $x + y$
2. Statement: performs an action: $z = x + y$.

What is a Statement in programming language?

Definition: A statement is an instruction that a program executes to perform some action.

Example: $z = x + y$

Explanation: This is an assignment statement.

It tells the program to:

1. Take the value of x,
2. Add it to the value of y
3. Store the result in a variable z

So, if $x = 2$, $y = 3$ then after execution $z = 5$

Check Your Knowledge:

1. Write a statement to subtract b from a and store in c.
2. Identify expression part and statement part:

- i) $x + 5$
- ii) $x = 5$
- iii) $a * b - 10$
- iv) `print("Hello")`
- v) $y = x * 2$
- vi) $10/2$
- vii) `if x > 5:`
`print(x)`
- viii) `len("python")`
- ix) $z = (x + y) * 2$

Answers:

- 1. $C = a - b$
- 2.
 - iv) $x + 5$
Ans: Expression
 - v) $x = 5$
Ans: Statement
 - vi) $a * b - 10$
Ans: Expression
 - vii) `print("Hello")`
Ans: Statement
 - viii) $y = x * 2$
Ans: Whole line statement, $x * 2$ part is expression.

vi) $10/2$

Ans: expression

vii) `if x > 5:`
`print(x)`

Ans: Statement.

Viii) `len("python")`
Ans: Expression

ix) $z = (x + y) * 2$

Ans: Whole Line is statement; $(x + y) * 2$ this part is expression

2. Arithmetic Operators:

Definition: Arithmetic operators are symbols used in programming language to perform mathematical calculations on variables and values.

Common Arithmetic Operators:

Operators	Meaning	Example	Output
+	Addition	5+ 2	7
-	Substraction	5 -2	3
*	Multiplication	5*2	10
/	Divison	5/2	2.5
//	Floor Division	5//2	2
%	Modulus (remainder)	5%2	1
**	Power	2**3	8

Practice:

1. Find the result: $a = 10, b = 3$; $\rightarrow a + b = ?$
2. Find the result: $a = 17, b = 5$; $\rightarrow a - b = ?$
3. Find the result: $x = 20, y = 6$; $\rightarrow x/y$
4. Find the output: $x = 8, y=3$, $\rightarrow x//y$
5. Find the output: $x = 3, y = 2$ $\rightarrow 3\%2$
6. Find the output: $x = 4, y = 2$ $\rightarrow x**y$

3. Logical Operators:

Definitions: Logical operators are used in programming to combine multiple conditions or change their true value, and they return either True or False.

Used for conditions → return True/False

Operators	Meaning
and	Both must be True
or	At least one True
not	reverse

Practice:

1. X = 5;

```
print(x>3 and x<10)  # True
print(x> 10 or x == 5) # True
print(not(x>3))      # False
```

2. Find the output: x = 10, y = 5 → (x>y) and (y>0).

3. Find the output: x = 3, y = 7 → (x>y) or (y>5)

4. Find the output: x =4 → not(x>2)

4. Comparison Operators:

Definition: Comparison operators are used in programming to compare two values and return a result of True or False.

Operator	Meaning
= =	Equal
!=	Not Equal
>	Greater
<	Less
>=	Greater than equal
<=	Less than equal

Practice:

1. Find the output: $x = 5, y = 5 \rightarrow x == y = ?$
2. Find the output: $x = 10, y = 7 \rightarrow x != y = ?$
3. Find the output: $x = 10, y = 7 \rightarrow x > y = ?$
4. Find the output: $x = 4, y = 2 \rightarrow x < y = ?$
5. Find the output: $x = 4, y = 2 \rightarrow x >= y = ?$
6. Find the output: $x = 4, y = 2 \rightarrow x <= y = ?$

5. Type Conversion:

Definition: Conversion of **data types** means changing a value from one data type to another.

What are the data types?

Definition: A data type tells us what kind of value a variable is storing.

Simple Understanding:

Data type = type of data.

(e.g., number, text, true/false)

Main data types in Python:

1. Integer(int)

- Whole numbers (no decimal)
- Example: `x = 10`.

2. Float(float)

- Numbers with decimal
- Example: `x = 3.14`

3. String(str)

- Text(written in quotes)
- Example: `name = "Pranjal"`

4. Boolean(bool)

- Only two values: True or False
- Example: `x = True`

5. List(list)

- Collection of multiple values (ordered, changeable)
- Example: `numbers = [1,2,3,4]`

6. Tuple(tuple)

- Collection of values (ordered, but not changeable)
- Example: numbers = (1,2,3)

7. Dictionary(dict)

- Stores data key-value pairs
- Example: person = {"name": "Pranjal", "age": 25}

8. Set(set)

- Unordered collection of unique values
- Example: s = {1,2,3}

Practice:

1. Identify the data type:

- i) 100
- ii) 4.5
- iii) "100"
- iv) True
- v) ["pranjal",2,3,4]
- vi) ("alibag", "iig", "4")
- vii) {"months":["jan", "Feb", "March"],
"year": [2025,2026]}
- viii) {"red", "blue", "white"}

Conversion of Data Types into Other data type:

Definition: Conversion of one data type into another data type is called type conversion or type casting.

Types of Conversion:

1. Implicit Conversion (Automatic)

- Done automatically by the programming language
- No user involvement

Example:

1. $x = 5$
2. $y = 2.5$
3. $z = x + y$

2. Explicit Conversion (Manual)

Done by the programmer using functions.

Example:

1. $x = \text{"10"} \rightarrow \text{int}(x) \rightarrow 10$

2. $x = 5.7$
 $y = \text{int}(x)$
 $\text{print}(y)$

3. $x = \text{"3.14"}$
 $y = \text{float}(x)$
 $\text{print}(y) \# 3.14$

Practice:

1. Integer Conversion

```
x = "10"  
y = int(x) # string → integer  
print(y)
```

2. Float Conversion

```
x = "3.14"  
y = float(x) # string → float  
print(y)
```

3. String Conversion

```
x = 100,  
y = str(x) # integer → string  
print(y)
```

4. Tuple Conversion

```
x = [1,2,3]  
y = tuple(x) # list → tuple  
print(y)
```

5. Set Conversion

```
x = [1,2,3,4,5]  
y = set(x) # list → set (remove duplicates)
```

6. Dictionary Conversion

```
x = [("a",1),("b",2)]  
y = dict(x) # list of tuples → dictionary  
print(y)
```

6. f- Strings

Definition: f-string is a way to insert variables or expressions directly inside a string.

Basic syntax:

f“your text {variable_or_expression}

Practice:

1. Name = “Pranjal”, Age = 35
print(f “My name is {name} and age is {age}”)

2. **Expression Inside:**
A = 5,
B = 3,
Print(f “Sum = {a+b}”)

3. **Formatting:**
Pi = 3.14159
Print(f “{pi:2f}”) # 3.14

7. Operator Precedence

Definition: Operator precedence in programming refers to the order in which operations are performed in an expression when multiple operators are present.

Like BODMAS

Order of precedence:

1. Paranthesis
2. Exponentiation
3. Multiplication/Division/Floor Division/ Modulus: These have same priority evaluated from left to right.
4. Addition/Substraction

$() \rightarrow ** \rightarrow * / // \% \rightarrow + -$

Practice:

1. $20/5 * 2 = ?$
2. $17//3*2 = ?$
3. $2 + 3*4 = ?$
4. $(2+3)*4 = ?$
5. $2+3*4=?$

7. Debugging

Definition: Debugging is the process of finding and fixing errors (bugs) in a program so that it runs correctly.

Debugging = finding and fixing errors.

Example:

Code with error:

```
a = 10  
b = 0  
print(a / b) # Error: Division by zero
```

Types of Errors:

Type 1: Syntax Error

Definition: Mistakes in writing a code (grammar rules of the language)

```
Print("Hello # Missing quote # Syntax Error
```

Type 2: Runtime Errors

Occur while the program is running.

1. $X = 10/0$
2. $X = \text{int}(\text{"abc"})$ # crash

Types of Runtime error:

i) Name Error

Definition: Variable is used before being defined.

```
Print(x)
```

ii) TypeError

Definition: Operation applied to wrong data type.

Example: `5 + "10"`

iii) ValueError

Definition: Correct type but invalid value.

Example: `int("abc")`

iv) IndexError

Definition: Accessing invalid index.

Example:

`List = [1,2],`

`Print(list[5])`

v) KeyError

Definition: Accessing non-existing dictionary key.

Example:

`d = {"a": 1}`

`print(d["b"])`

vi) ZeroDivisionError

Definition: Division by zero

Example:

`10/0`

vii) AttributeError

Definition: Object has no such method/attribute.

Example:

`x = 10`

`x.append(5)`

viii) FileNotFoundError

Definition: File does not exist.

Example: `open("file.txt")`

ix) ImportError/ModuleNotFoundError

Definition: Module cannot be imported.

Example:

`import xyz`

Type 3: Logical Error

Program run, but gives wrong output.

Example: `result = 5 - 3`

`A = 10;`

`B = 5;`

`Print(a - b) # You wanted addition but wrote sub-traction`

Debugging Techniques

Simple Strategy for debugging:

1. Read error message.
2. Check line number
3. Print variables
4. Simplify code
5. Use debugger (if needed: VS code, PyCharm)

6. Use print()

`A = 10;`

`B = 5;`

`Print("a=",a)`

`Print("b=",b)`

`Print(a+b)`

7. Use try-except

Try:

```
X = int("abc")
```

Except:

```
Print("Conversion failed")
```

8. Use VS Code Debugger

a) Add breakpoint , b) Run → Debug, c) Check variable step by step.